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# C Programming Project Presentation

- Project Title: [Matrix Operations Tool]
- Course: C Programming Lab / Mini Project
- Department: Computer Science


Team 4

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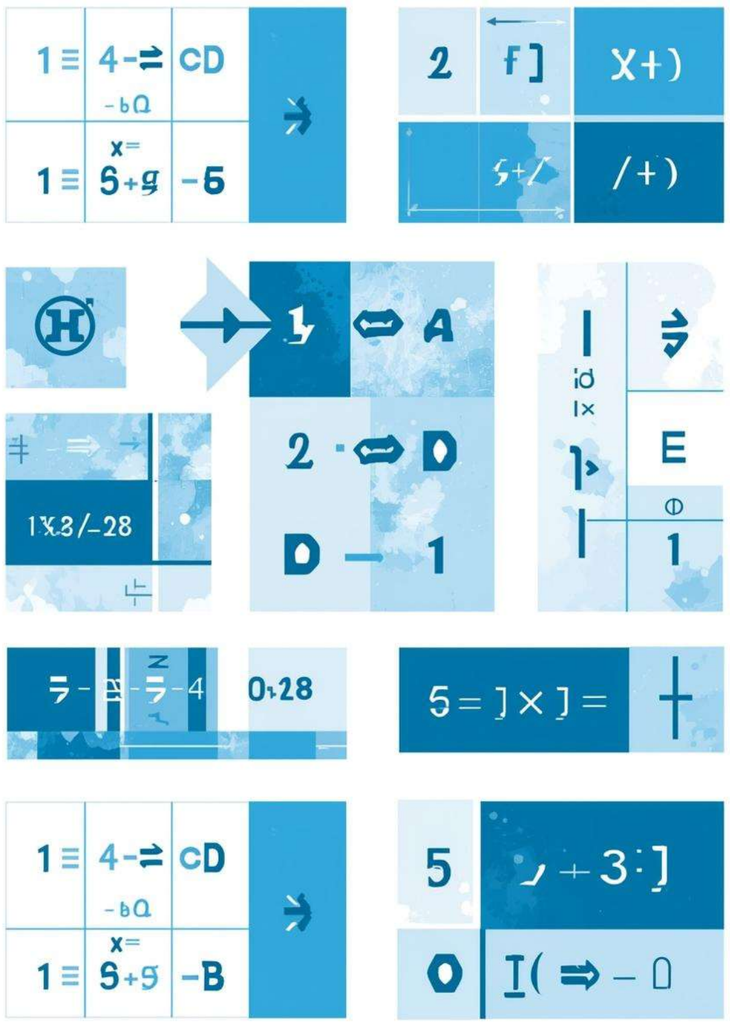
# *Introduction*

## The Importance of Automating Matrix Calculations in C Programming

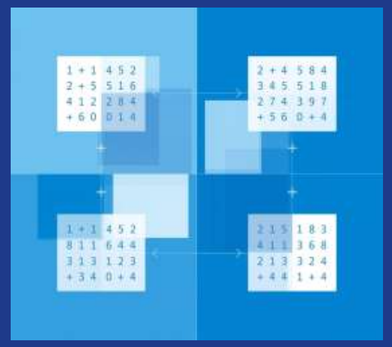
Automating matrix computations significantly enhances efficiency and accuracy in mathematical operations. This project demonstrates how to leverage C programming to facilitate essential matrix functions such as addition, subtraction, multiplication, and transposition. A user-friendly interface ensures that users can perform these operations seamlessly, reducing manual errors and saving time.



# Project Objectives

## Automation

**Streamline calculations** for efficiency and accuracy



## Programming

Utilize arrays and **control structures** effectively

## User Interface

Design a **friendly menu** for navigation

# System Requirements

## Essential Hardware and Software for Optimal Performance

To effectively run the Matrix Operations Tool, your system should meet these minimum requirements: an **Intel i3 processor or higher**, at least **4GB of RAM**, and an operating system of either **Windows or Linux**. Additionally, the tool needs a **GCC compiler** to execute the C programming code seamlessly.

# Step-by-Step Algorithm Overview

1 Start program

2 Display menu:

1 → Add 2 → Subtract 3 → Multiply

4 → Transpose 5 → Row Sum 6 → Column Sum 7 → Exit

3 Read user choice

4 If choice = 7: Display "Exit" → Stop

5 If Add/Sub:

- Read rows & cols (r, c)
- Read matrices A & B
- Perform A + B
- Display result

6 If Multiply:

- Read sizes $A(r_1, c_1)$, $B(r_2, c_2)$
- If $c_1 \neq r_2$ → print "Can't multiply"
- Else multiply A × B → Display result

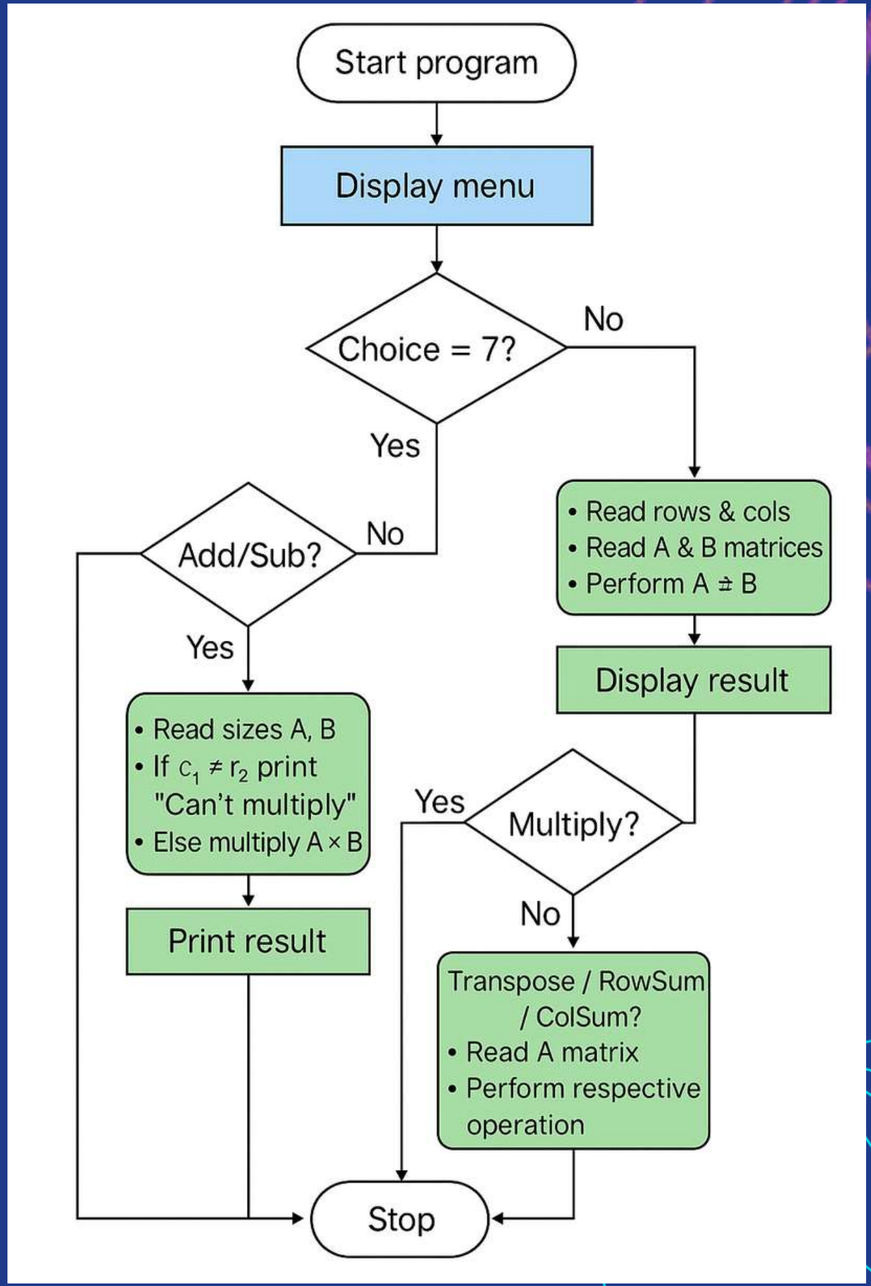
7 If Transpose / RowSum / ColSum:

- Read matrix A
- Perform respective operation → Display result

8 Return to menu and repeat

# Flowchart Design

Visual representation of the system's workflow steps.



# *Module Description*

## Key Components for Matrix Operations and User Interaction

The **Matrix Operations Tool** consists of three main modules: the Input Module gathers user data, the Processing Module executes calculations, and the Output Module presents results clearly. This modular design promotes clarity, enhances user experience, and maintains the program's functionality, allowing for easy updates and improvements in the future.

### Input Module

Phose for sllrr g leccoune chateg!

### Processing Module

Phose for sturre lescoune chateg!

### Output Module

Startl manketing escounte ditersiu.

## ⚙️ Implementation of Code Logic — Code Structure

**Modular Design:**

Separate functions handle specific tasks for clarity and reusability.

**Main Components:**

main() → Menu, user input, operation selection

readMatrix(), printMatrix() → Input / Output

addMatrices(), subMatrices(), mulMatrices() → Core math logic

transposeMatrix(), sumOfRows(), sumOfColumns() → Extra tools

**Flow:**

Menu → Read Choice → Validate → Perform Operation → Display Result → Repeat

Data Used:

2D arrays A, B, R (size ≤ MAX × MAX)

Key Feature:

Clean modular structure, simple logic, easy maintenance

# Sample Output of Matrix Operations

```
===== MATRIX OPERATIONS TOOL =====

Choose an operation:
1 - Addition (A + B)
2 - Subtraction (A - B)
3 - Multiplication (A x B)
4 - Transpose (of A)
5 - Sum of each Row (of A)
6 - Sum of each Column (of A)
7 - Exit
Enter your choice (1-7):
```

```
===== MATRIX OPERATIONS TOOL =====

Input Matrix A:
Enter rows and columns for Matrix A (rows cols): 2 2
Enter elements of Matrix A (2 x 2):
1 2
1 3

Now input Matrix B:
Enter rows and columns for Matrix B (rows cols): 2 2
Enter elements of Matrix B (2 x 2):
2 3
2 3

Operations Menu (using stored A and B):
1  - Add (A + B)
2  - Subtract (A - B)
3  - Multiply (A x B)
4  - Transpose (A)
5  - Sum of each Row (A)
6  - Sum of each Column (A)
7  - Show stored matrices (A and B)
8  - Replace Matrix A
9  - Replace Matrix B
0  - Exit
Enter your choice: 3
Result (A x B):
6   9
8   12
```

# *Results*

## Evaluating the Effectiveness of the Matrix Operations Tool

The Matrix Operations Tool has demonstrated **accurate results** and efficient processing capabilities. Users can perform basic matrix computations interactively, showcasing the importance of programming logic and user-friendly design. This tool not only enhances computational speed but also reduces human error, making it a valuable asset for academic purposes.

# *Conclusion & Future Enhancements*

## Reflecting on achievements and potential advancements in matrix operations

This project has successfully achieved its goal of **automating matrix calculations**, enhancing both speed and accuracy. Looking ahead, future enhancements may include the addition of determinant and inverse calculations, as well as the ability to save results to files, further broadening the tool's functionality.

# *References & Acknowledgements*

## Resources that supported the development of the matrix operations tool

**This project benefited from a variety of resources including textbooks, online tutorials, and guidance from mentors. Special thanks to my teammates for their collaboration and insights, which were invaluable in refining the design and implementation of the matrix operations tool.**

# Thank you